

OFFICIAL TECHNICAL & BUSINESS SPECIFICATION

AI WhatsApp Automation & Multi-Tenant CRM Platform

An autonomous client engagement, appointment booking, and customer lifecycle management system. Built specifically for clinics, wellness centers, high-touch agencies, and high-volume service businesses.

 Live Production: crm.goboldlabs.com Multi-Tenant Cryptographic Isolation Multi-LLM BYOK Engine 2-Way Google Calendar Sync Meta WhatsApp Cloud API

Key Business Outcomes & Quantifiable ROI

Deploying this system replaces fragmented manual communication, missed phone calls, and manual spreadsheets with an autonomous 24/7 intelligent engine.

< 3 Sec**INSTANT RESPONSE TIME**

Autonomous 24/7 replies to WhatsApp leads while interest is peak. Never lose a lead to competitor delay.

45% – 60%**BOOKING CONVERSION LIFT**

Conversational qualification guides prospects directly to confirmed Google Calendar slots.

70% Drop**NO-SHOW REDUCTION**

Automated 24-hour and 2-hour WhatsApp reminders with 1-click reschedule nudges.

15 – 25 Hrs**STAFF TIME SAVED / WEEK**

Eliminates repetitive inquiry answering, manual slot scheduling, and follow-up logging.

Before vs. After Implementation

OPERATIONAL AREA	TRADITIONAL MANUAL PROCESS	BOLDLABS AI WHATSAPP AUTOMATION SYSTEM
Inbound Lead Response	Delayed 30 mins to 4 hours. Enquiries outside business hours remain unanswered until morning. High lead abandonment.	Instant <3s Response 24/7: AI answers inquiries immediately with accurate business info, services, and doctor profiles.
Appointment Booking	Back-and-forth messaging: "What time works for you?" Staff manually cross-checks calendars, causing double bookings and friction.	Autonomous Slot Locking: AI checks live Google Calendar availability, proposes exact open slots, and locks appointments instantly.
No-Shows & Cancellations	15% to 35% no-show rate. Staff forgets manual reminder calls; cancelled slots remain empty and generate zero revenue.	Zero-Friction Reminders: Automated WhatsApp alerts with instant reschedule links. Released slots immediately reopen for new bookings.
Follow-ups & Retention	Unconverted prospects and past clients are forgotten. No systematic re-engagement mechanism exists.	Automated CRM Follow-Up Engine: Scheduled nudges, post-service review requests, and personalized broadcast re-engagement campaigns.
Team Collaboration	Single physical smartphone passed around front desk; no visibility into who replied or customer history.	Unified Team CRM & Live Inbox: Multi-user web dashboard, role-based access, instant human takeover toggle, and customer timeline.

System Architecture: Core Modules Breakdown

The platform comprises seven tightly integrated sub-systems engineered for reliability, enterprise security, and maximum conversion velocity.

1. Autonomous AI WhatsApp Agent (BYOK)

AI CORE

Powered by state-of-the-art LLMs (Groq Llama 3.3 70B, Google Gemini 3.1 Flash-Lite, and OpenCode AI) with dedicated per-tenant prompt engineering.

- ✓ **Intent Recognition:** Automatically detects booking requests, reschedules, cancellations, pricing inquiries, and human takeover requests.
- ✓ **Brand Persona & Voice:** Fully customizable assistant name (e.g. *Dr. Sameer, Rakshaya*), tone (warm, clinical, consultative), and style.
- ✓ **Knowledge Base Grounding:** Ingests exact service catalogs, pricing tiers, doctor credentials, operating hours, and clinic directions.
- ✓ **Strict Guardrails:** Enforces methodology rules (e.g. consultative selling, objection handling) and prevents hallucinations.

Business Outcome: 95%+ of customer inquiries are resolved instantly without front-desk staff touching the keyboard.

2. 2-Way Google Calendar Sync Engine

CALENDAR

Direct OAuth2 integration with Google Calendar providing real-time slot verification, instant appointment booking, and automatic calendar cleanup.

- ✓ **Live Free/Busy Slot Check:** Evaluates real-time staff availability against business operating hours, slot duration, and existing entries.
- ✓ **Automated Calendar Event Creation:** Creates rich Google Calendar events with customer phone, service details, and notes.
- ✓ **Instant WhatsApp Confirmation:** Dispatches approved Meta HSM confirmation template with calendar invite immediately upon booking.
- ✓ **Automated Slot Release:** When a client cancels or reschedules, the calendar event is deleted/updated instantly, freeing the slot for others.

Business Outcome: Zero double-bookings; front desk schedules update in real time on mobile phones and desktops automatically.

3. Omnichannel CRM & Live Team Inbox

LIVE CRM

A centralized, real-time web portal that gives your entire team immediate visibility into every active customer conversation and booking status.

- ✓ **Real-Time Message Sync:** Bi-directional chat inbox with instant status indicators (sent, delivered, read receipts).
- ✓ **1-Click AI Pause / Human Takeover:** Toggle AI control per customer with a single click when personalized human attention is needed.
- ✓ **Customer Profile Drawer:** Instant side-panel showing lead status, total appointments, lifetime spend, notes, and tags.
- ✓ **Pre-Approved Meta Template Dispatcher:** Send official outbound HSM templates directly from the chat window at any time.

Business Outcome: Smooth collaboration across doctors, front desk, and sales staff with zero risk of duplicated or conflicting messages.

4. Public Self-Service Booking Pages

CLIENT PORTAL

Each client organization gets dedicated, lightning-fast public booking URLs (e.g. `crm.goboldlabs.com/[slug]/book`) customized to their brand.

- ✓ **Custom Service Catalog:** Displays treatments, consultation fees, session durations, and description cards.
- ✓ **Live Interactive Slot Picker:** Dynamically calculates open appointment times across any date based on live Google Calendar data.
- ✓ **Client Details Collection:** Captures patient name, WhatsApp phone number, age, location, and primary health concern.
- ✓ **Instagram / Google Ads Ready:** Clean link to paste in Instagram bios, Google My Business listings, and paid ad landing pages.

Business Outcome: Drives direct appointments from social media and search ads straight into your calendar 24 hours a day.

System Architecture: Core Modules Breakdown (Cont.)

Advanced retention marketing, automated staff alerting, enterprise security, and multi-tenant management.

5. Customer Pipeline & Follow-Up Automation

PIPELINE

Transforms raw phone numbers into a structured CRM pipeline with intelligent lead scoring and automated task management.

- ✓ **Lifecycle Stages:** Categorizes contacts across *New, Contacted, Consulted, Converted, Retained, and Inactive*.
- ✓ **Lead Probability Scoring:** Visual indicators (*Cold, Warm, Hot*) for prioritized front-desk follow-up.
- ✓ **Customer Clinical Notes Board:** Color-coded internal notes (author tagged) to record medical observations, special instructions, and client preferences.
- ✓ **Google Tasks Synchronization:** Automated follow-up task creation with deadlines that sync directly to Google Tasks on staff devices.

Business Outcome: No lead slips through the cracks; staff always knows who to follow up with and what treatment plan was discussed.

6. Marketing Hub & Re-Engagement Engine

MARKETING

Re-activates dormant customers and collects 5-star Google reviews systematically using Meta-approved WhatsApp broadcasting.

- ✓ **Automated Re-Engagement Triggers:**
 - ✓ 3-Day Unconfirmed Nudge (for inquiries that haven't booked)
 - ✓ 7-Day Post-Consultation Wellness Check-in
 - ✓ 30-Day Re-activation Reminder for periodic follow-ups
- ✓ **Automated Review Generation:** Automatically sends your Google Review or Instagram feedback link 2 hours post-appointment.
- ✓ **Meta Template Catalog Sync:** Fetches all approved Meta templates and displays live status directly in the CRM dashboard.
- ✓ **Campaign Broadcasts:** Bulk broadcast promotional offers or announcements to segmented customer lists.

Business Outcome: Increases repeat appointment bookings by 30%+ and organically drives 5-star reviews on Google and Instagram.

7. Multi-Channel Staff Alerts & Digest

ALERTS

Ensures doctors and front-desk personnel receive instant notifications across every critical touchpoint without logging into the software.

- ✓ **Instant Staff WhatsApp Alerts:** Doctor/receptionist phone receives instant WhatsApp notification template when an appointment is booked.
- ✓ **Direct Gmail Notifications:** Beautiful HTML email receipts sent immediately with patient name, contact, selected service, and time.
- ✓ **Web Push Notifications (VAPID):** Native browser alerts on desktop and mobile even when the CRM tab is closed.
- ✓ **Automated Morning Digest:** Daily executive summary outlining today's schedule, patient count, and open follow-up tasks.

Business Outcome: Zero missed appointments; staff is instantly prepared for walk-ins and consultations well in advance.

8. Enterprise Multi-Tenancy & Security

SECURITY

Engineered from the ground up as an enterprise multi-tenant platform with absolute cryptographic separation of data and credentials.

- ✓ **Strict Data Isolation:** Every database query, contact, chat, and appointment is filtered by tenant UUID. Zero cross-tenant data leakage.
- ✓ **Independent Credentials:** Each client uses their own Meta WhatsApp Phone ID, Google Calendar token, and AI API keys.
- ✓ **Super Admin Master Portal (/admin/clients):** Centralized management to onboard clients, set monthly plans, and reset passwords.
- ✓ **Granular Role-Based Access (RBAC):** Specific permission sets for *Admin, Doctor, Receptionist, Sales, and Viewer*.

Business Outcome: Enterprise-grade security and client data privacy compliance (HIPAA / GDPR friendly).

End-to-End Customer Journey Blueprint

Here is how an inbound customer seamlessly travels through the system from initial WhatsApp message to repeat customer:

1

Inbound WhatsApp Inquiry Received (<3s)
Prospect messages your business number. Meta webhook delivers payload to Redis queue. Core worker processes intent and context.

2

Autonomous AI Consultative Reply & Slot Suggestion
AI agent answers queries accurately using your clinic's knowledge base and checks Google Calendar for genuine open appointment slots.

3

Appointment Confirmed & Google Calendar Locked
Prospect confirms slot. System locks the slot, creates Google Calendar event, updates CRM pipeline to *Converted*, and notifies staff.

4

Instant Multi-Channel Staff & Customer Confirmation
Patient receives WhatsApp confirmation with Google Calendar event. Staff receives instant WhatsApp alert + HTML email notification.

5

Automated Reminders (24h & 2h Prior) with Reschedule Protection
Scheduled job triggers WhatsApp reminder template. If patient needs to reschedule, AI handles the time change or provides a 1-click self-service link.

6

Post-Visit Review Collection & Automated Re-engagement
2 hours post-session, a review request template is sent. At 7 and 30 days, re-engagement triggers check in on patient progress to encourage re-booking.

What Every Onboarded Client Receives (Turnkey Deliverables)

DELIVERABLE	DESCRIPTION & SPECIFICATIONS	DEPLOYMENT STATUS
Dedicated CRM Workspace	Branded web portal at <code>crm.goboldlabs.com/[client_slug]</code> with company logo, custom terminology, timezone, and currency.	READY & LIVE
Official Meta Cloud API	Official WhatsApp Business Cloud API setup with green-badge readiness, zero risk of phone ban, and high throughput.	FULLY INTEGRATED
Trained AI Assistant Persona	Custom knowledge base with your doctor profile, treatment menu, pricing, clinic address, and objection handling rules.	CUSTOM CONFIGURED
Google Calendar 2-Way Sync	Live free/busy slot checking, automated event creation, and slot release on cancellation or reschedule.	ACTIVE SYNC
Public Booking Portal	Mobile-optimized booking link (<code>/[slug]/book</code>) for Instagram bio, Google My Business, and ad campaigns.	LIVE PRODUCTION
Staff Alert Channels	Instant WhatsApp alerts to staff phone, rich Gmail booking receipts, and real-time browser desktop push notifications.	ACTIVE ALERTS
Multi-User Staff Logins	Individual logins for doctors, front desk, and marketing with customized permission gates.	RBAC CONFIGURED

Platform Production Status & Infrastructure Health
Hosted on dedicated high-performance cloud architecture (Oracle Cloud Infrastructure) with redundant Docker containers, PostgreSQL 15, Redis stream queues, and Nginx reverse proxy. Validated with **89/89 automated system tests (100% pass rate)** verifying multi-tenant isolation, Google Calendar sync, WhatsApp messaging, and API security.